

BoT-IoT MAQT Training Setups

This note summarizes how lawun chose MAQT training settings on FROZEN BoT-IoT, and how those runs compare to arjun's reported CE-head test accuracy.

Latest (full known train, cosine start 0.01): about 68% CE-head accuracy on the full known test with no reject mask — up from the previous coreset pick (3000 / class, same LR recipe, about 65%). See section 7.

Unless stated otherwise, confusion matrices below are Algo-3 accepted-known test results: known-class test rows with nonconformity $s(x) \leq q$ (Global conformal). Rejected known-test rows are excluded from accuracy and from the matrix.

1. Baseline: 4-class, 3000 / class, fixed LR 0.05

Setup: four known classes (DDoS, DoS, Normal, Reconnaissance); DPP coreset capped at 3000 per class; fixed learning rate 0.05.

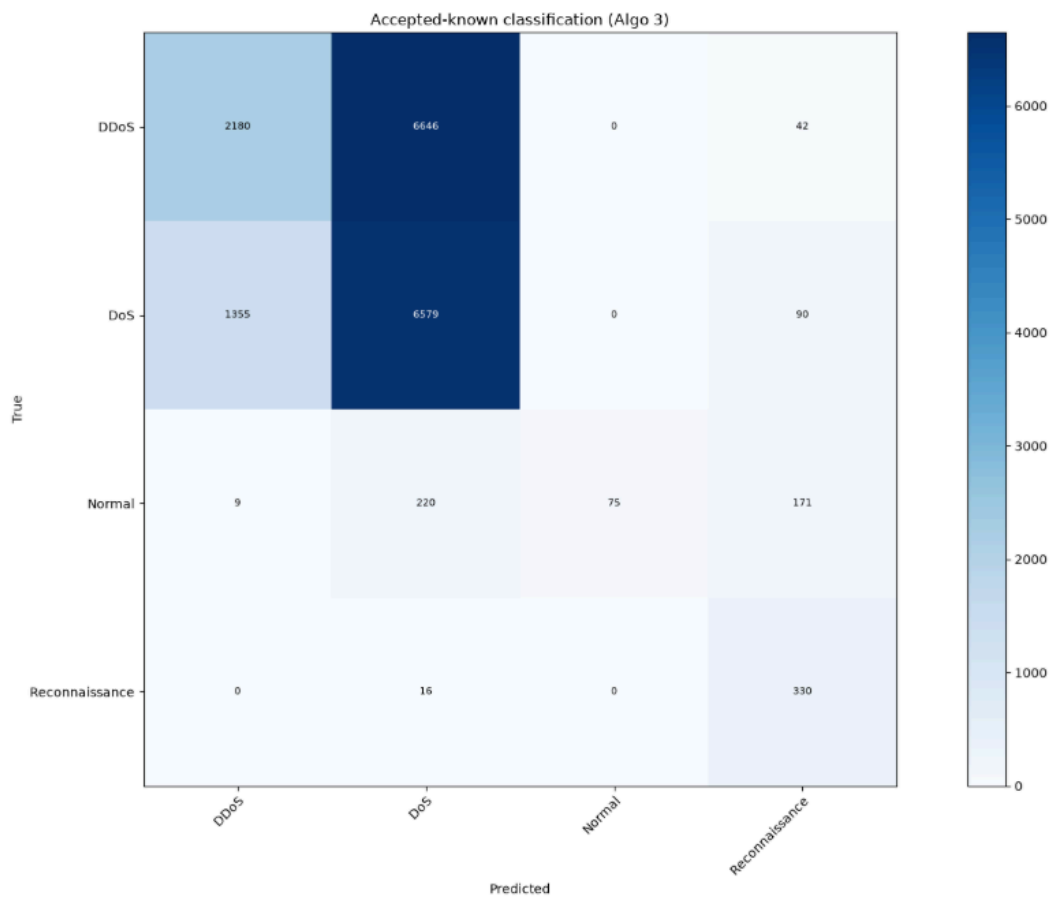
Observation: overall accepted accuracy is modest (~52%). Normal is poorly recovered (high precision, very low recall). DDoS and DoS are heavily confused.

```
[test] mean certified radius (accepted): 0.0125
[test] reject rate (known): 0.0472 | acc (accepted): 0.5174

[test] classification report (accepted only):
      precision    recall  f1-score   support

   DDoS         0.62     0.25     0.35     8868
    DoS         0.49     0.82     0.61     8024
   Normal        1.00     0.16     0.27       475
 Reconnaissance  0.52     0.95     0.67       346

 accuracy         0.66
 macro avg         0.54
 weighted avg      0.57
```



2. Ablation: drop DoS (3-class), same budget

Hypothesis: majority DoS / DDoS traffic might be starving Normal.

Setup: train without DoS (DDoS + Normal + Reconnaissance only); still 3000 / class, LR 0.05.

Observation: Normal does not improve (accepted Normal recall stays at zero in this run). High headline accuracy is driven by DDoS dominance, not better Normal geometry.

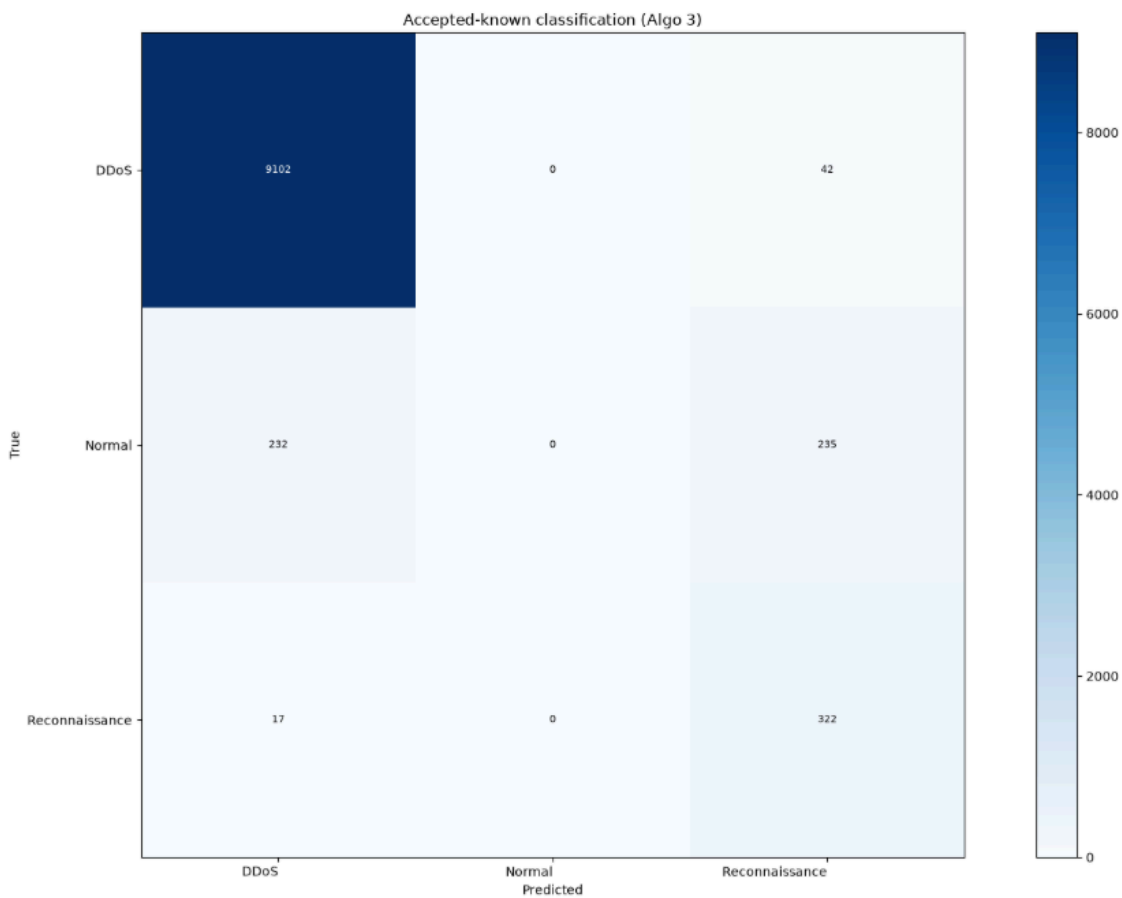
Conclusion: DoS presence is not the main cause of poor Normal prediction → return to 4-class.

```
[test] mean certified radius (accepted): 0.0710
[test] reject rate (known): 0.0484 | acc (accepted): 0.9471

[test] classification report (accepted only):
      precision    recall  f1-score   support

   DDoS         0.97      1.00      0.98      9144
  Normal         0.00      0.00      0.00       467
Reconnaissance    0.54      0.95      0.69       339

 accuracy         0.95      0.95      0.95      9950
 macro avg        0.50      0.65      0.56      9950
 weighted avg     0.91      0.95      0.93      9950
```



3. Scale data: 4-class, 20000 / class, LR 0.05

Setup: larger DPP coreset (20000 per class), same fixed LR 0.05.

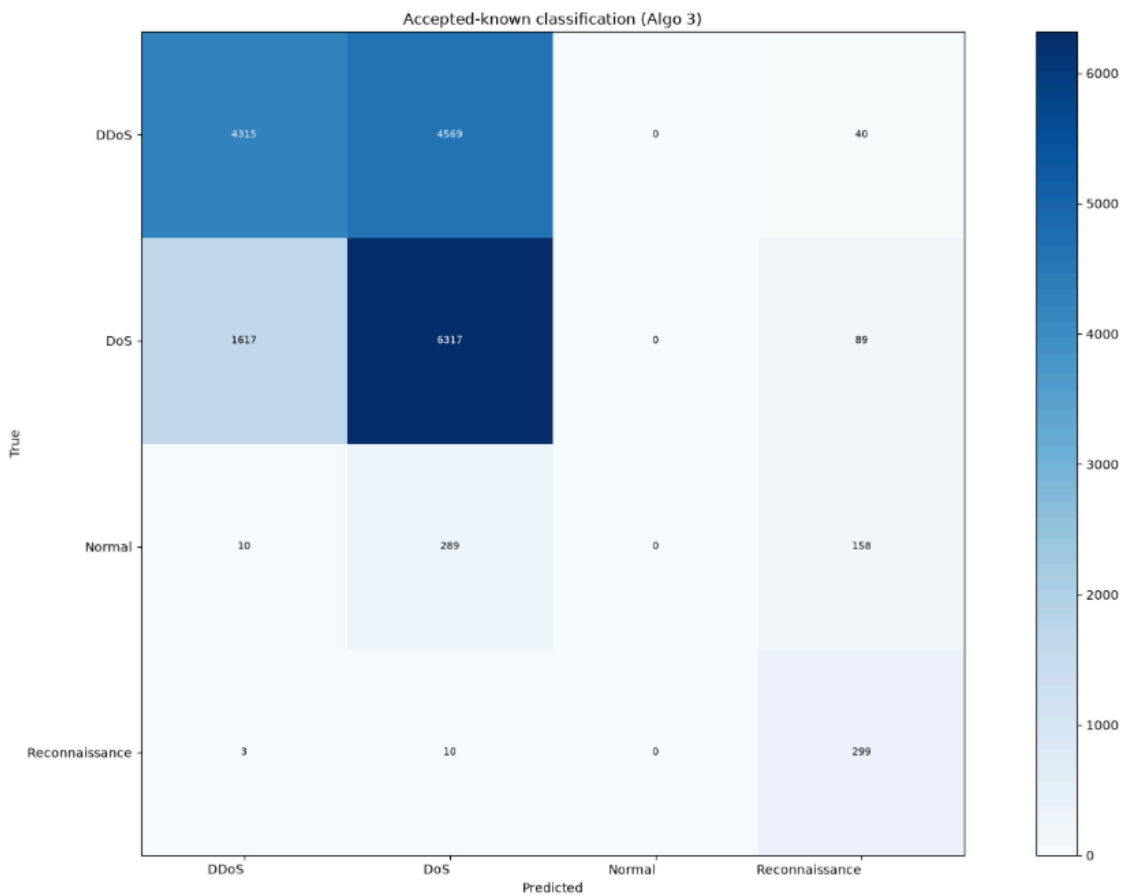
Observation: accepted accuracy rises (~62% vs ~52% at 3000). DDoS ↔ DoS confusion remains large; Normal stays weak (near-zero correct in this matrix). More data helps overall accuracy but does not fix class separation.

```
[test] mean certified radius (accepted): 0.0082
[test] reject rate (known): 0.0471 | acc (accepted): 0.6170

[test] classification report (accepted only):
      precision    recall  f1-score   support

   DDoS         0.73     0.48     0.58     8924
    DoS         0.56     0.79     0.66     8023
   Normal         0.00     0.00     0.00         457
 Reconnaissance 0.51     0.96     0.67        312

 accuracy         0.45
 macro avg         0.56
 weighted avg         0.62
```



4. Smaller fixed LR, moderate size: 10000 / class, LR 0.0005

Setup: 10000 per class, fixed LR 0.0005.

Observation: accepted accuracy (~53%) is close to the 3000 / LR 0.05 baseline (only a small bump, on the order of about one percentage point in this comparison). DDoS ↔ DoS and weak Normal persist. Extra samples + tiny fixed LR do not justify the cost for this pipeline.

```

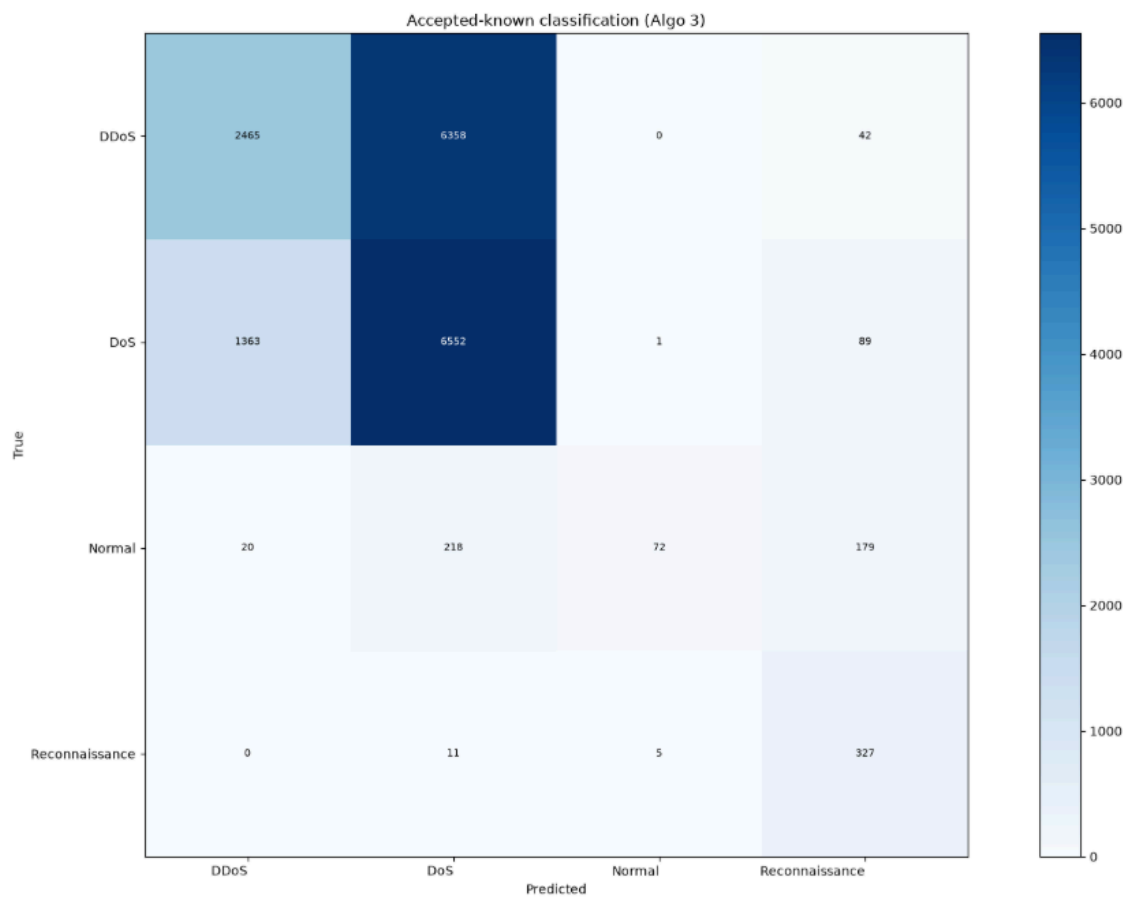
[test] mean certified radius (accepted): 0.0132
[test] reject rate (known): 0.0478 | acc (accepted): 0.5319

[test] classification report (accepted only):
      precision    recall  f1-score   support

   DDoS         0.64      0.28      0.39      8865
    DoS         0.50      0.82      0.62      8005
   Normal        0.92      0.15      0.25       489
 Reconnaissance  0.51      0.95      0.67       343

 accuracy         0.64      0.55      0.53      17702
  macro avg         0.64      0.55      0.48      17702
  weighted avg         0.58      0.53      0.49      17702

```



5. Back to 3000 / class with cosine LR schedule

Given diminishing returns from larger coresets under fixed LR, lawun returns to 3000 per class and turns on automatic LR decay (`use_lr_schedule=True`, cosine annealing from the starting LR toward a small floor).

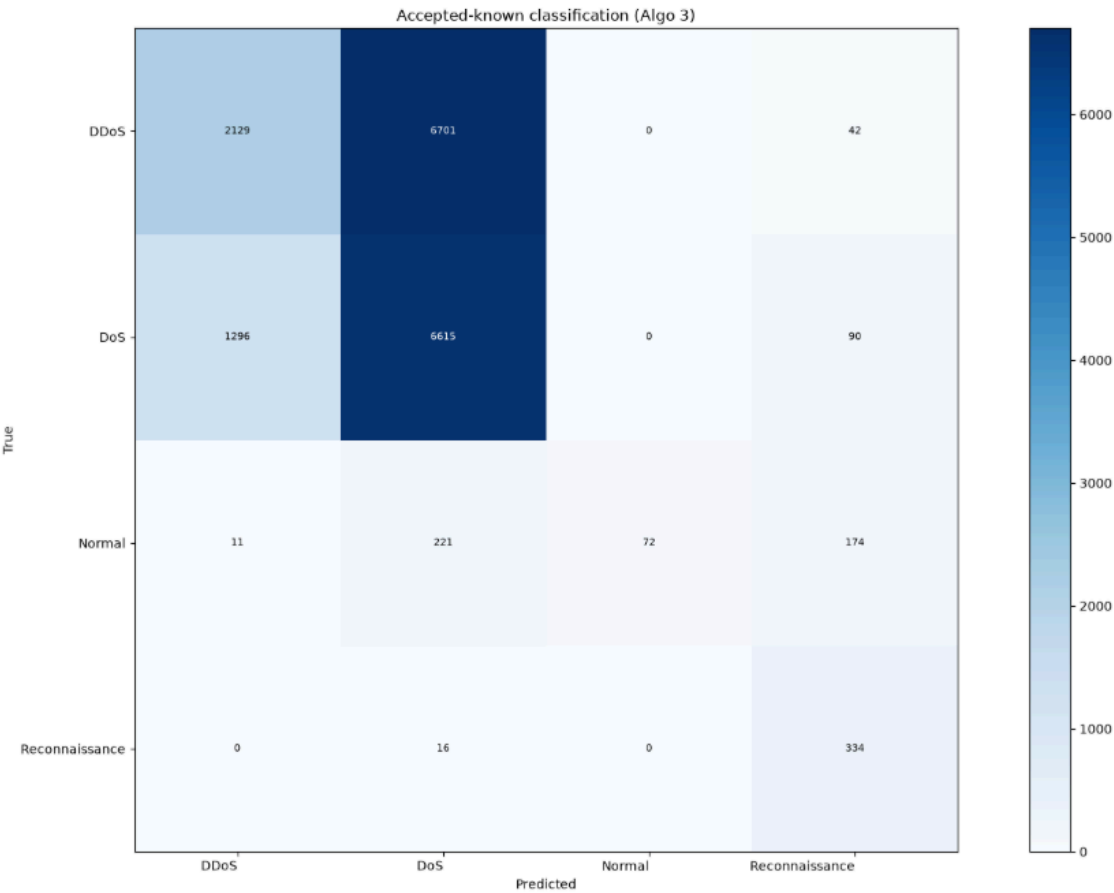
5a. Start LR 0.03

```
[test] mean certified radius (accepted): 0.0136
[test] reject rate (known): 0.0479 | acc (accepted): 0.5169

[test] classification report (accepted only):
      precision    recall  f1-score   support

   DDoS           0.62       0.24       0.35       8872
    DoS           0.49       0.83       0.61       8001
   Normal          1.00       0.15       0.26        478
Reconnaissance     0.52       0.95       0.67        350

 accuracy           0.66
  macro avg          0.54
 weighted avg        0.52
```



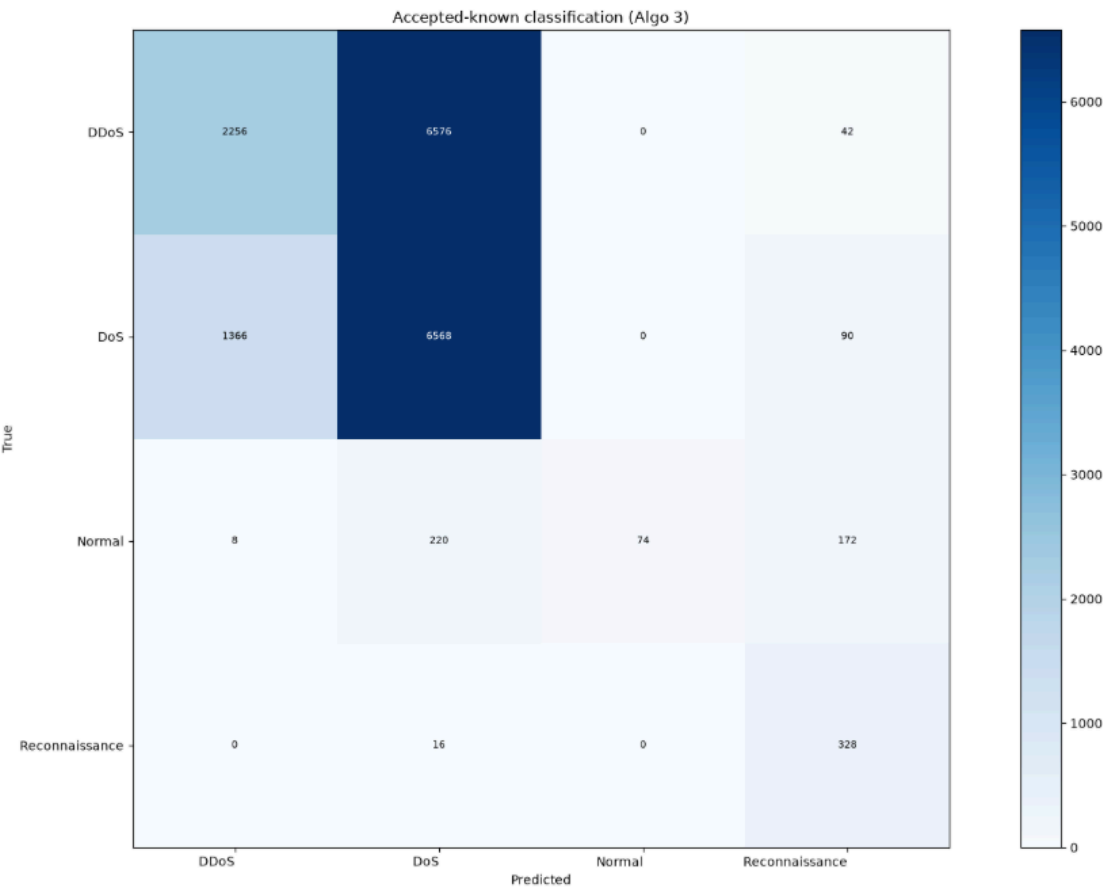
5b. Start LR 0.01 (chosen)

```
[test] mean certified radius (accepted): 0.0120
[test] reject rate (known): 0.0471 | acc (accepted): 0.5208

[test] classification report (accepted only):
      precision    recall  f1-score   support

   DDoS         0.62      0.25      0.36      8874
    DoS         0.49      0.82      0.61     8024
   Normal        1.00      0.16      0.27       474
Reconnaissance    0.52      0.95      0.67       344

 accuracy         0.66
  macro avg         0.55
 weighted avg         0.52
```



Observation: accepted metrics for start 0.03 vs 0.01 are similar (~52%). Training with start 0.01 is judged smoother on the loss/LR curves, so the coreset-era working choice was:

Knob	Choice
Classes	4 (DDoS, DoS, Normal, Reconnaissance)
Train coreset	3000 / class (DPP)
LR	cosine schedule, start 0.01
Notebooks	4class-3000each-lr0.01auto-maqt → 4class-3000each-lr0.01auto-algo3 (+ cached variants)

This was later superseded by full-train under the same LR recipe (section 7).

6. Fair compare to arjun (full known test, no reject mask)

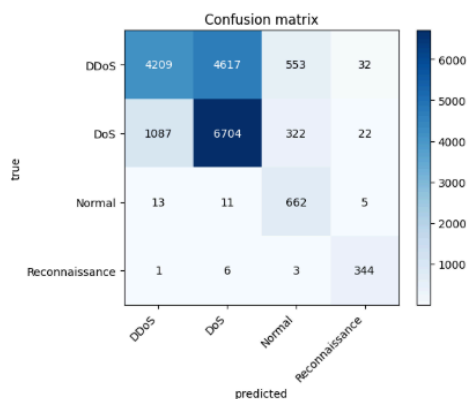
Algo-3 accepted-only accuracy is not comparable to a CE-head score on every known-test row.

- arjun (ablation-test-bot-iot): CE-head `predict_labels` on full known test → about 64% accuracy ($n = 18591$).
- lawun best MAQT (4-class, 3000 / class, cosine start 0.01): CE-head on the same full known test, no conformal mask → about 65% accuracy ($n = 18591$).

arjun (~64%, full known test, CE, no reject)

test accuracy: 0.6411

	precision	recall	f1-score	support
DDoS	0.79	0.45	0.57	9411
DoS	0.59	0.82	0.69	8135
Normal	0.43	0.96	0.59	691
Reconnaissance	0.85	0.97	0.91	354
accuracy			0.64	18591
macro avg	0.67	0.80	0.69	18591
weighted avg	0.69	0.64	0.63	18591

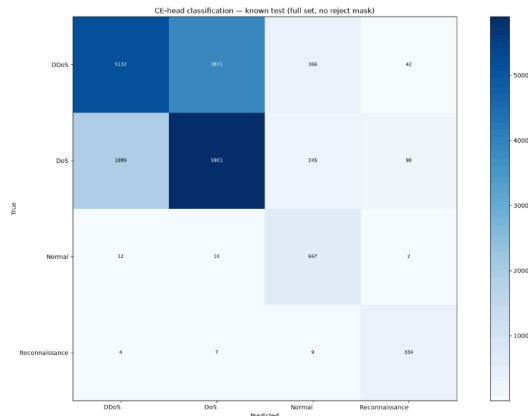


lawun (~65%, full known test, CE, no reject)

```
(test CE) accuracy (full set, no reject mask): 0.6473
(test CE) macro-F1 (full set): 0.6823
(test CE) n=3000

(test CE) classification report (full set):
precision recall f1-score support
DDoS 0.73 0.55 0.62 9411
DoS 0.68 0.73 0.68 8135
Normal 0.52 0.97 0.67 691
Reconnaissance 0.71 0.94 0.81 354

accuracy 0.64 0.79 0.65 18591
macro avg 0.67 0.80 0.68 18591
weighted avg 0.67 0.65 0.64 18591
```



Under that matched protocol, the 3000 / class run was on par with (slightly above) arjun's reported CE full-test accuracy. Remaining open issue in those coresets: DDoS vs DoS separation; conformal accepted-only CMs still showed weak Normal recall when rejects are stripped out.

7. Latest: full train (~148k), cosine start 0.01 → ~68% CE full-test

Motivation: the section 5 and 6 working choice (3000 / class + cosine 0.01, ~65% CE full-test) left headroom if the model saw all FROZEN known-train rows instead of a DPP coreset.

Setup: 4-class; `SUBSET=False` (full known train, ~148k); cosine LR schedule, start 0.01; batch 64; $\lambda_1 = 0.5$, $\lambda_2 = 0.1$; same FROZEN features / circuit recipe as `4class-all-lr0.01auto-lambda2-maqt`.

Protocol (matched to section 6): CE-head on full known test, no conformal reject mask ($n = 18591$).

Metric	3000 / class, cosine 0.01 (section 6)	Full train, cosine 0.01 (this)
Accuracy	~65%	~68.2%
Macro-F1	—	~0.73
n	18591	18591

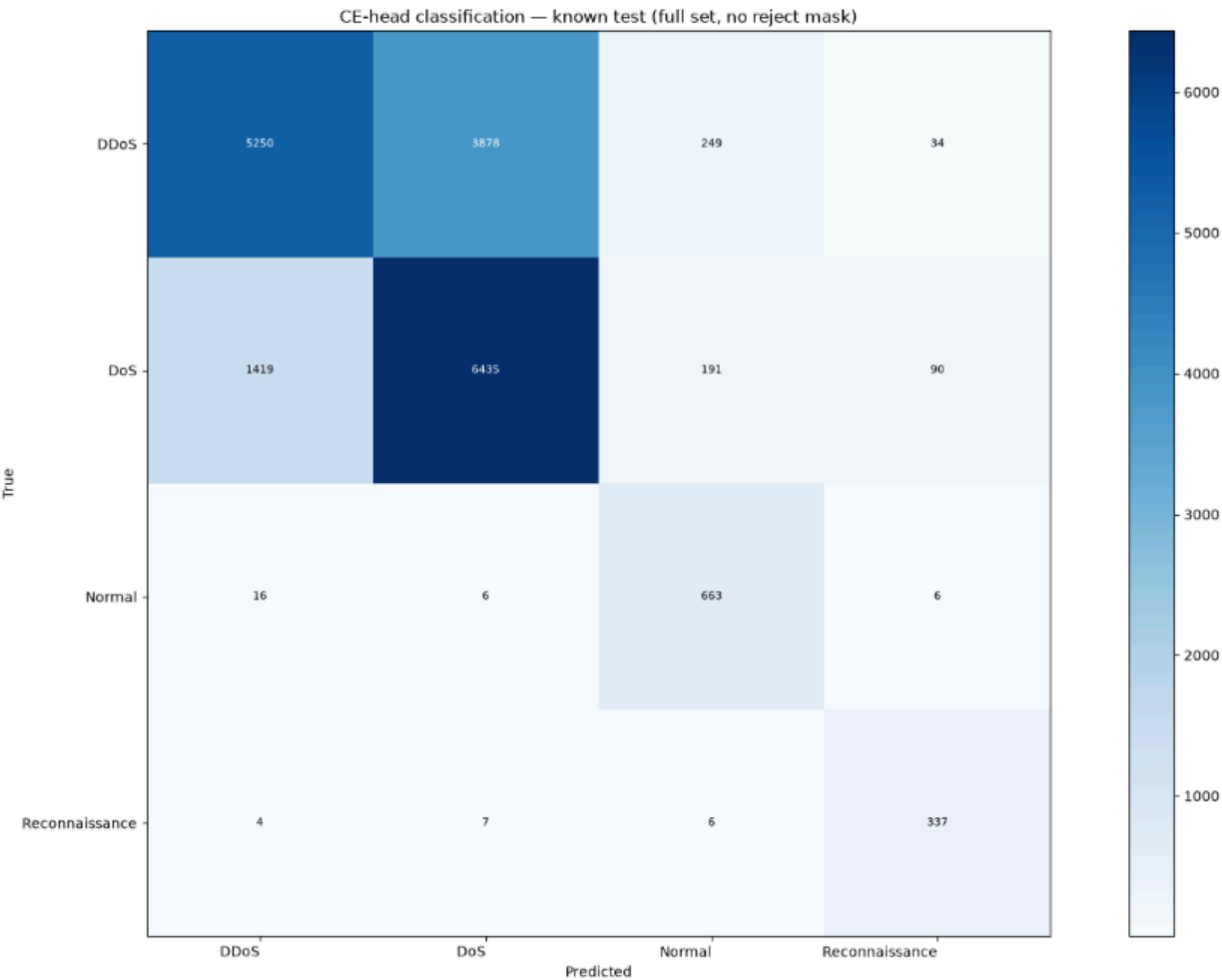
Observation: more train samples lift CE full-test accuracy by about +3 points vs the previous 3000-coreset run. Normal and Reconnaissance look strong on this matrix (high recall); the main residual error is still DDoS ↔ DoS swap.


```
[test CE] accuracy (full set, no reject mask): 0.6823
[test CE] macro-F1 (full set): 0.7267
[test CE] n=18591

[test CE] classification report (full set):
      precision    recall  f1-score   support

   DDoS         0.78     0.56     0.65     9411
    DoS         0.62     0.79     0.70     8135
   Normal        0.60     0.96     0.74      691
Reconnaissance    0.72     0.95     0.82      354

 accuracy         0.68     0.68     0.68    18591
  macro avg         0.68     0.82     0.73    18591
 weighted avg         0.71     0.68     0.68    18591
```



Current working choice:

Knob	Choice
Classes	4 (DDoS, DoS, Normal, Reconnaissance)
Train data	full FROZEN known train (no per-class cap)
LR	cosine schedule, start 0.01
Notebooks / artifacts	4class-all-lr0.01auto-lambda2-maqt (+ algo3 / cached variants)

Decision trail (short)

```
4c / 3k / lr=0.05
  $→$ Normal weak, DDoS↔DoS confused
3c (drop DoS) / 3k / lr=0.05
  $→$ Normal still not fixed $→$ back to 4c
```

```
4c / 20k / lr=0.05
  $\rightarrow$ accuracy up; DDoS↔DoS / Normal still hard
4c / 10k / lr=0.0005
  $\rightarrow$  $\approx$  3k result; not worth the extra data for this goal
4c / 3k / cosine 0.03 vs 0.01
  $\rightarrow$ similar accepted metrics; prefer 0.01 (smoother)
previous latest: 4c / 3k / cosine start 0.01 (~65% CE full-test vs arjun
~64%)
current latest: 4c / full train / cosine start 0.01
  $\rightarrow$ ~68% CE full-test (no reject); Normal/Recon strong;
DDoS↔DoS still open
```